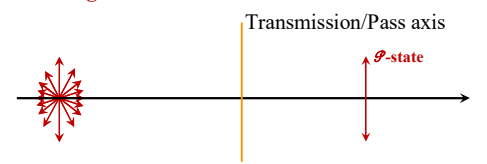


Polarization

Production of Polarized light



Transmission/Pass axis

P-state

- ❁ Dichroism (Selective Absorption)
- ❁ Birefringence (Double refraction)
- ❁ Reflection
- ❁ Scattering

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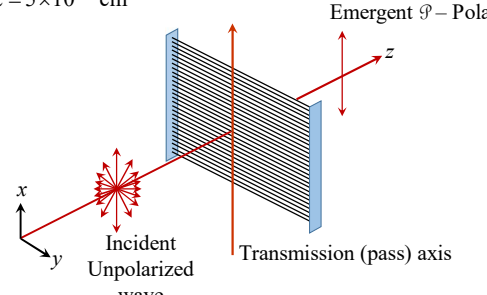
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Polarization

Production of Polarized light: Dichroism

For polarizer to be effective: $d \leq \lambda$

- ➡ For microwave $\lambda = 3 \text{ cm}$
- ➡ For lightwave $\lambda = 5 \times 10^{-5} \text{ cm}$



Incident Unpolarized wave

Transmission (pass) axis

Emergent P - Polarized light

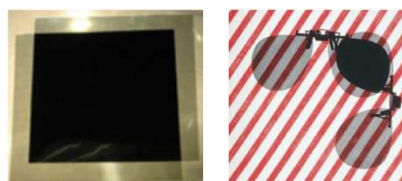
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Polarization

Production of Polarized light: Polaroid

- **Edwin Herbert Land:** (19 year old UG student at Harvard College) in 1928 invented the first dichroic sheet polarizer.
- Molecular analog of the wire grid polarizer.
- Iodine attached with the aligned chain of long polymeric molecules of PVA acts like thin long conducting wire.
- Component of $\vec{E}$ parallel to the molecular chain gets absorbed.



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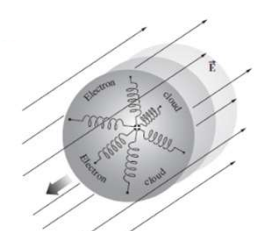
50

Polarization

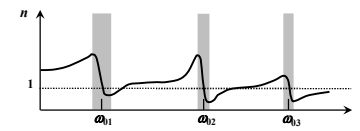
Production of Polarized light: Birefringence (Double refraction)

- All springs are the same with spring constant k
- Medium possesses unique *natural/resonant* frequency:

$$\omega_0 = \sqrt{k/m_e}$$

$$n^2(\omega) = \frac{\epsilon(\omega)}{\epsilon_0} = 1 + \frac{Nq_e^2}{\epsilon_0 m_e} \sum_i \frac{f_i}{\omega_{0i}^2 - \omega^2}$$


Mechanical oscillator model for an isotropic medium

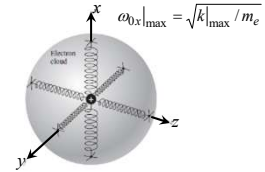


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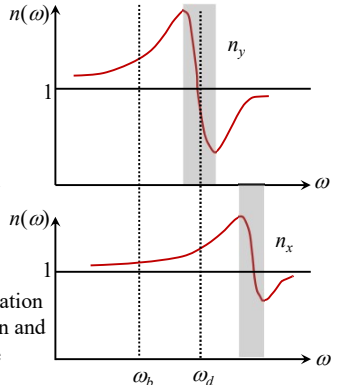
Polarization

Production of Polarized light: Birefringence (Double refraction)



■ Springs in different directions have different k
 ➔ Medium possesses three *natural (resonant)* frequencies in the three directions!

■ In an **anisotropic medium**, velocity of propagation in general depends on the propagation direction and the SOP ➔ Double refraction/Birefringence

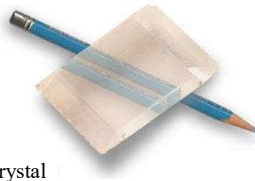
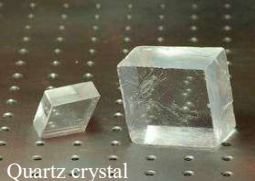


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Polarization

Production of Polarized light: Birefringence (Double refraction)

Calcite crystal Quartz crystal

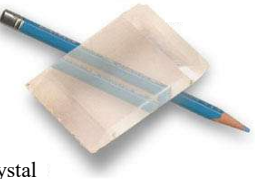
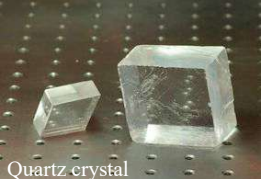
■ Birefringent materials display two different (direction dependent) refractive indices.

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Calcite crystal Quartz crystal

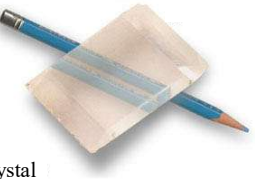
■ Ray entering the crystal splits up into two rays:
 these rays propagate along different directions with different velocities
 they are linearly and mutually orthogonally polarized
 one of the ray follows Snell's law of refraction → **Ordinary ray (o-ray)**
 other ray does not follow Snell's law of refraction → **Extraordinary Ray (e-ray)**

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Polarization

Production of Polarized light: Birefringence (Double refraction)


Calcite crystal Quartz crystal

■ Birefringent materials display two different (direction dependent) refractive indices.
 ■ In such **anisotropic medium**, velocity of propagation in general depends on the propagation direction and the SOP ➔ Double refraction/Birefringence
 ■ There used to be specific directions, along which velocities of o-ray and e-ray are same
 ➔ Optic axis ■ Uniaxial medium/biaxial medium

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Polarization

Production of Polarized light: Birefringence (Double refraction)

-ve uniaxial crystal
 $(n_o > n_e)$
(Calcite)
 $n_o = 1.6584$
 $n_e = 1.4864$

+ve uniaxial crystal
 $(n_o < n_e)$
(Quartz)
 $n_o = 1.5443$
 $n_e = 1.5534$

o-ray velocity: $v_o = c/n_o$
 e-ray velocity: $\frac{1}{v_e^2} = \frac{n_e^2(\theta)}{c^2} = \frac{\cos^2 \theta}{c^2/n_o^2} + \frac{\sin^2 \theta}{c^2/n_e^2}$

→ e-ray travelling $\parallel$ optic axis ($\theta = 0^\circ$) $v_e = c/n_o$
 → e-ray travelling $\perp$ optic axis ($\theta = 90^\circ$) $v_e = c/n_e$

o-ray: $\vec{E}, \vec{D} \perp$ Optic axis, $\vec{k}$
 e-ray: $\vec{E} \perp \vec{E}$ of o-ray, $\vec{k}$

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Calcite crystal (-ve)

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Calcite crystal (-ve) : $(n_o > n_e)$

Optic axis $\perp$
 $\theta = 0$
 $\theta = \pi/2$
 $v_o = c/n_o$
 $v_e = c/n_e$

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Calcite crystal (-ve) : $(n_o > n_e)$

Optic axis $\perp$
 $\theta = \pi/2$
 $v_o = c/n_o$
 $v_e = c/n_e$

Oblique incidence
 $n_i \sin \theta_i = n_o \sin \theta_r$ $n_i \sin \theta_i = n_e \sin \theta_r$

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Polarization

Production of Polarized light: Birefringence (Double refraction)
 Calcite crystal (-ve) : ($n_o > n_e$)

Wollaston Prism

Rochon Prism

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Retarders/Retardation plates
 Class of optical elements, which serves to change SOP of the incident wave.

- ✿ Phase shift suffered by o-ray: $\phi_o = k_0 n_o l$
- ✿ Phase shift suffered by e-ray: $\phi_e = k_0 n_e l$
- ✿ Crystal introduces a phase difference:

$$\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l$$
- ✿ Quarter wave plate: $\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l = \frac{\pi}{2}$
- ✿ Half wave plate: $\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l = \pi$
- ✿ For **Calcite**, $\vec{E} \parallel$ optic axis constitutes e-ray and travels faster

➔ direction of optic axis in -ve uniaxial retarder → **FAST axis**

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Retarders/Retardation plates
 Class of optical elements, which serves to change SOP of the incident wave.

- ✿ Phase shift suffered by o-ray: $\phi_o = k_0 n_o l$
- ✿ Phase shift suffered by e-ray: $\phi_e = k_0 n_e l$
- ✿ Crystal introduces a phase difference:

$$\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l$$
- ✿ Quarter wave plate: $\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l = \frac{\pi}{2}$
- ✿ Half wave plate: $\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l = \pi$
- ✿ For **Calcite**, $\vec{E} \perp$ optic axis constitutes o-ray and travels slower

➔ direction $\perp$ optic axis (FAST axis) in -ve uniaxial retarder → **SLOW axis**

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Retarders/Retardation plates
 Class of optical elements, which serves to change SOP of the incident wave.

- ✿ Phase shift suffered by o-ray: $\phi_o = k_0 n_o l$
- ✿ Phase shift suffered by e-ray: $\phi_e = k_0 n_e l$
- ✿ Crystal introduces a phase difference:

$$\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l$$
- ✿ Quarter wave plate: $\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l = \frac{\pi}{2}$
- ✿ Half wave plate: $\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l = \pi$
- ✿ For **Quartz**, $\vec{E} \perp$ optic axis constitutes o-ray and travels faster

➔ direction $\perp$ optic axis in +ve uniaxial retarder → **FAST axis**

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Retarders/Retardation plates
Class of optical elements, which serves to change SOP of the incident wave.

- Phase shift suffered by o-ray: $\phi_o = k_0 n_o l$
- Phase shift suffered by e-ray: $\phi_e = k_0 n_e l$
- Crystal introduces a phase difference:
$$\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l$$
- Quarter wave plate: $\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l = \frac{\pi}{2}$
- Half wave plate: $\Delta\phi = \frac{2\pi}{\lambda_0} |n_o - n_e| l = \pi$
- For **Quartz**, $\vec{E} \parallel$ optic axis constitutes e-ray and travels slower
 direction of optic axis in +ve uniaxial retarder → **SLOW axis**

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Retarders/Retardation plates

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Polarization

Production of Polarized light: Birefringence (Double refraction)

Retarders/Retardation plates

Incident light, linearly polarized $\parallel$ to either principal axis (Slow/Fast) will be unaffected by the retarder.

A relative phase difference can only be introduced by the retarder if the incident field has finite component along the two mutually orthogonal directions of principal axes.

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Polarization

Polarization (General case)

Action of Quarter-wave-plate

Action of Half-wave-plate

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Polarization

Jones Matrix: Application

Example 1: Unpolarized light passes through a LP (pass axis @ 45° from horizontal). Light emerging from the LP falls on $\lambda/4$ -waveplate (of Quartz) whose optic axis is vertical. Find the output SOP.

Solution: (a) SOP after LP
 $E_x = E_0 \cos 45^\circ \cos(kz - \omega t)$
 $E_y = E_0 \sin 45^\circ \cos(kz - \omega t)$

(b) SOP input to $\lambda/4$ -waveplate
 $E_x = \frac{E_0}{\sqrt{2}} \cos(kz - \omega t)$
 $E_y = \frac{E_0}{\sqrt{2}} \cos(kz - \omega t)$

(c) Identifying SA & FA of QWP
 Quartz $\lambda/4$ -waveplate $\rightarrow +ve$ uniaxial crystal
 $\rightarrow v_o > v_e \rightarrow n_o < n_e$
 Optic axis VERICAL
 E_x leads E_y
 o-ray (fast) e-ray (slow)

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Polarization

Jones Matrix: Application

Example 1: Unpolarized light passes through a LP (pass axis @ 45° from horizontal). Light emerging from the LP falls on $\lambda/4$ -waveplate (of Quartz) whose optic axis is vertical. Find the output SOP.

Solution: (d) Field after passing through QWP
 $E_x = \frac{E_0}{\sqrt{2}} \cos(kz - \omega t)$
 $E_y = \frac{E_0}{\sqrt{2}} \cos\left(kz - \omega t + \frac{\pi}{2}\right)$
 $= -\frac{E_0}{\sqrt{2}} \sin(kz - \omega t)$

$\rightarrow$ If origin is brought at the plane next to the QWP, then at this new plane having $z = 0$,
 $E_x = \frac{E_0}{\sqrt{2}} \cos \omega t$
 $E_y = \frac{E_0}{\sqrt{2}} \sin \omega t$

$\rightarrow$ LCP
 CCW rotation

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Polarization

Jones Matrix: Application

Example 2: Unpolarized light passes through a LP (pass axis @ 45° from horizontal). Light emerging from the LP falls on two $\lambda/4$ -waveplates (of Quartz) whose optic axis is horizontal. Find the output SOP.

Solution:

$\rightarrow$ ρ -state at -45°

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